FACULTY OF COMPUTING

SEMESTER 2

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SECP2753-02 DATA MINING

**PROJECT 2**


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## TABLE OF CONTENT

**SECTION A**

## 1.0 INTRODUCTION

This project uses unsupervised learning techniques to discover meaningful patterns within a diabetes dataset by clustering patients based on clinically relevant variables. The goal is to group individuals into segments that reflect shared diabetes characteristics, allowing for improved understanding of risk factors and more targeted intervention strategies. Clustering helps uncover hidden structures in the data without relying on predefined labels, which is especially useful in exploratory.

In particular, this project explores several business insights derived from relationships among variables such as BMI, Cholesterol_Total, Glucose_Fasting, Blood_Pressure_Diastolic, and HbA1c. These insights are clinically important for identifying patients at increased risk of metabolic syndrome, cardiovascular conditions, and diabetes.

To support these insights, two clustering algorithms were implemented which are K-Means Clustering Algorithm and Agglomerative Hierarchical Clustering (AGNES). These methods were used to segment the data into groups of similar patient profiles. The resulting clusters provide actionable information for healthcare providers, such as identifying individuals who may benefit from early lifestyle intervention, additional screening, or more intensive medical follow up.

## 2.0 CLUSTERING ALGORITHM

In this project, we applied a clustering algorithm to discover natural groupings within the data. These methods help identify underlying patterns without using predefined labels. Two clustering that we used are:

1. K-Means Clustering : Partitions data into k groups based on distance to the cluster centroid. This method assumes spherical clusters and works best with well separated data.
2. Agglomerative Hierarchical Clustering: Builds a nested tree (dendrogram) of clusters by iteratively merging the closest clusters using Ward's linkage.

## 3.0 BUSINESS INSIGHT

1. Higher BMI is associated with higher Cholesterol_Total levels.
   Patients with higher BMI tend to have increased cholesterol levels, indicating a possible link between excess body weight and dyslipidemia risk.

   Dataset Characteristics:

   - The dataset includes numeric columns for both BMI and Cholesterol_Total. This allows direct correlation analysis between body mass and lipid health indicators, showing that heavier patients tend to have higher cholesterol.

2. Patients with higher Glucose_Fasting levels often have higher Blood_Pressure_Diastolic.
   Those with elevated fasting glucose commonly show increased diastolic pressure too in which to spotlight dual risks for metabolic syndrome.

   Dataset Characteristics:
   - Both Glucose_Fasting and Blood_Pressure_Diastolic are continuous numeric columns. This allows joint analysis of glucose metabolism and blood pressure regulation, useful for identifying metabolic syndrome tendencies in the population.

3. Patients with higher HbA1c values tend to have higher Glucose_Fasting levels.
   Fasting glucose rises alongside HbA1c, affirming their clinical relationship and potential to cross-predict each other in diabetes screening.

   Dataset Characteristics:

- Both HbA1c and Glucose_Fasting are numeric lab test results recorded for each patient. Their relationship confirms the dataset accurately captures blood glucose regulation metrics, enabling validation of clinical diabetes diagnostic standards.

### 3.1 Clinical Risk Segmentation : BMI and Cholesterol Analysis

In this project, unsupervised learning was applied to segment patients based on their body mass index (BMI) and total cholesterol levels. These two clinical features were selected due to their well established association with cardiometabolic risk. The goal of this segmentation is to uncover underlying patterns within the patient population that may reflect different levels of health risk. By identifying these groups, healthcare providers can develop more tailored intervention strategies such as prioritizing high-risk individuals for early lifestyle intervention or clinical monitoring while allocating fewer resources to those in lower risk categories. The insights derived from this clustering approach can support more targeted and efficient healthcare planning.

#### 3.1.1 Data Preprocessing

1. **Feature Selection**
   This analysis focuses on two clinically important variables which are Body Mass Index (BMI) and Cholesterol_Total. These features were selected due to their established relationship in medical literature, where excess body weight is often associated with dyslipidemia and increased cardiovascular risk.

```
df_bmi_chol = df[["BMI", "Cholesterol_Total"]].copy()
```

2. **Handling Missing Values**
   Mean imputation was applied to handle missing values. Each missing value was replaced with the mean of its respective column. This method was chosen because it is simple and effective, especially when the percentage of missing data is relatively low and randomly distributed.

```
imputer = SimpleImputer(strategy="mean")
df_bmi_chol_imputed = imputer.fit_transform(df_bmi_chol)
```

3. **Normalization**
   To ensure fair contribution of both variables in distance based clustering, z-score standardization was performed using StandardScaler. This transformation gives both BMI

and cholesterol values a mean of 0 and a standard deviation of 1, allowing meaningful Euclidean distance calculation.

```
scaler = StandardScaler()
df_bmi_chol_scaled = scaler.fit_transform(df_bmi_chol_imputed)
```

### 3.1.2 Model Evaluation

To determine the optimal number of clusters, the silhouette score was computed for different values of k. The highest score was achieved when k=2, indicating that two groups offer the best balance of separation and cohesion. These are relatively strong scores, indicating good cluster separation.

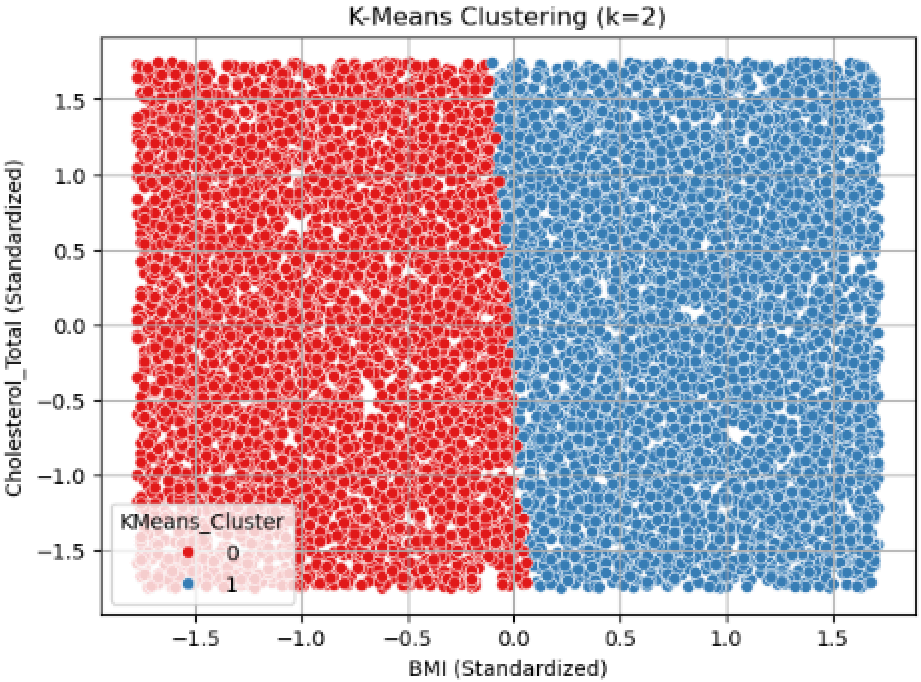
```
KMeans (k=2) Silhouette Score: 0.351
Agglomerative (k=2) Silhouette Score: 0.347
```

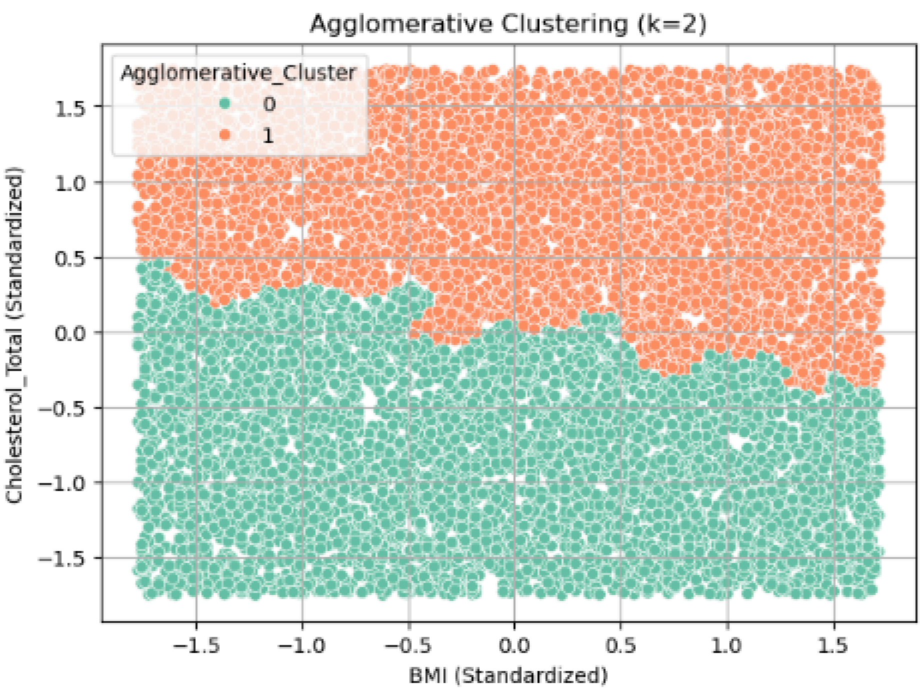
### 3.1.3 Visualization and Interpretation

Scatter plots of HbA1c versus Fasting Blood Glucose were used to visualize the clustering results. These two variables are key indicators in diabetes risk assessment, making them suitable for visual interpretation.

- In the K-Means clustering plot: the scatter plot of standardized BMI vs. Cholesterol_Total shows two distinct clusters, with nearly vertical boundaries. This indicated that BMI plays a stronger role in separating the clusters. One cluster represents individuals with higher BMI and likely higher cholesterol levels, reinforcing their clinical relationship.
- The Agglomerative clustering plot: this method also separates the data into two reasonably clear groups, although the boundary is less sharp than in K-Means. The separation follows a slightly curved trend, suggesting subtle variation in how cholesterol increases with BMI.
- The dendrogram from hierarchical clustering confirms a top level separation into two clusters, with multiple subgroupings. This tree-like visualization provides evidence of natural grouping and supports the use of k=2 in agglomerative clustering.

These visualizations and metrics confirm that BMI and cholesterol are strong candidates for patient segmentation.


K-Means Clustering (k=2)
Cholesterol_Total (Standardized)
BMI (Standardized)
KMeans_Cluster
0
1
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5


Agglomerative Clustering (k=2)
Cholesterol_Total (Standardized)
BMI (Standardized)
Agglomerative_Cluster
0
1
1.5
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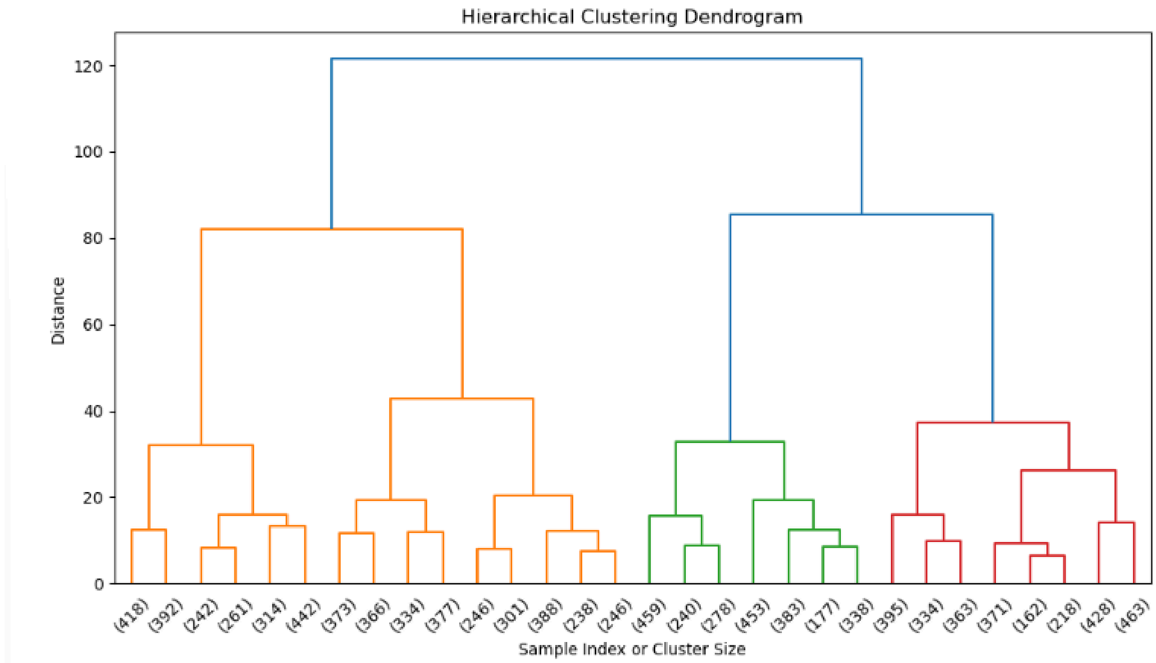
0.0
−0.5
−1.0
−1.5




### 3.1.4 Business Insight

This clustering analysis offers a valuable business insight for patients with higher BMI who tend to exhibit higher total cholesterol levels. These patients may be at increased risk of cardiovascular issues and may require early lifestyle intervention or pharmacological support.

By identifying this subgroup :

- Healthcare providers can target weight management programs for individuals in the high BMI/cholesterol cluster.
- Preventive campaigns may focus on diet and physical activity for this group to mitigate future health risks.
- Conversely, individuals in the lower BMI/cholesterol cluster can be managed through routine screening and wellness monitoring.

This segmentation allows for cost effective healthcare planning by prioritizing higher risk individuals for more intensive follow up while managing low risk individuals more efficiently.

### 3.2 Patients with higher Glucose_Fasting levels often have higher Blood_Pressure_Diastolic

Secondly, unsupervised learning techniques are also applied to segment patients based on their fasting blood glucose and diastolic blood pressure values. These two indicators were selected due to their strong relevance in identifying metabolic syndrome, a condition characterized by the co-occurrence of multiple cardiometabolic risk factors. The aim of this clustering analysis is to reveal distinct patient subgroups that may exhibit dual risks for elevated blood sugar and hypertension. By uncovering these patterns, healthcare providers can more effectively prioritize comprehensive screening and integrated management plans for individuals at heightened risk, supporting proactive and personalized healthcare delivery.

#### 3.2.1 Data Preprocessing

1. **Feature Selection**
   Glucose_Fasting and Blood_Pressure_Diastolic were selected as clustering features for this analysis. This allows for identifying patient groups with simultaneous risks of high blood sugar and elevated diastolic pressure.

```
df_agnes = df[['Fasting_Blood_Glucose', 'Blood_Pressure_Diastolic']].copy()
```

2. **Handling Missing Values**
   To ensure data quality before clustering, we checked for missing values in the dataset using `.isnull().sum()`. Any rows containing missing data were then removed with `.dropna(inplace=True)`.

```
print(df_agnes.isnull().sum())
df_agnes.dropna(inplace=True)
```

3. **Normalization**
   Before performing clustering, the dataset was normalized using `StandardScaler()` to standardize features to have a mean of 0 and a standard deviation of 1. This step ensures that all variables contribute equally to the clustering process by removing differences in scale.

```
scaler = StandardScaler()
scaled_features = scaler.fit_transform(df_agnes)
```

### 3.2.2 Model Evaluation

The silhouette score was calculated for various k values to identify the ideal number of clusters. By conducting silhouette scores, three clusters for K-Means and four clusters for AGNES show the optimal combination of separation and cohesion as the highest score obtained is when the k=3 and k=4 respectively. Although the result is not particularly strong, this analysis helps to identify a cluster structure that is acceptable for further analysis.

Silhouette score for both clusters algorithm:

```
Silhouette Score for K-Means clustering: 0.404
```

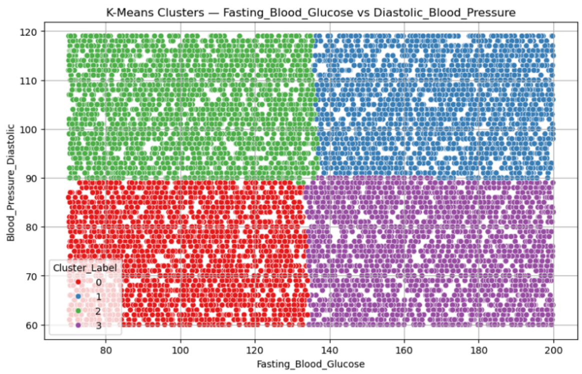
```
Silhouette Score for AGNES clustering: 0.320
```

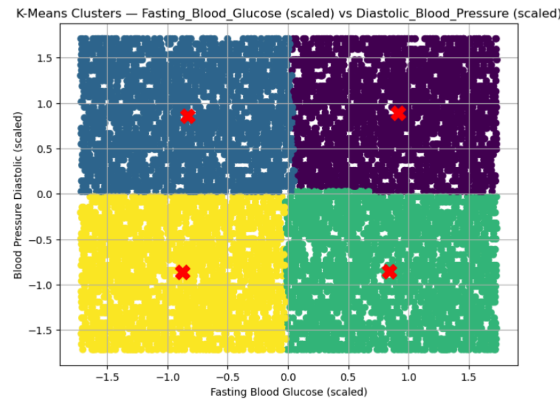
### 3.2.3 Visualization and Interpretation

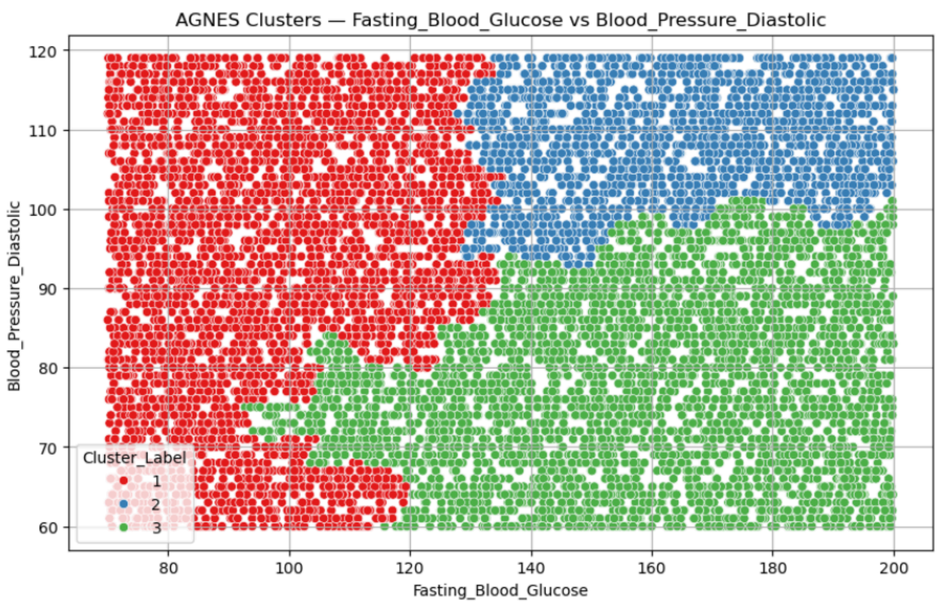
Scatter plots of HbA1c versus Fasting Blood Glucose were used to visualize the clustering results. These two variables are key indicators in diabetes risk assessment, making them suitable for visual interpretation.

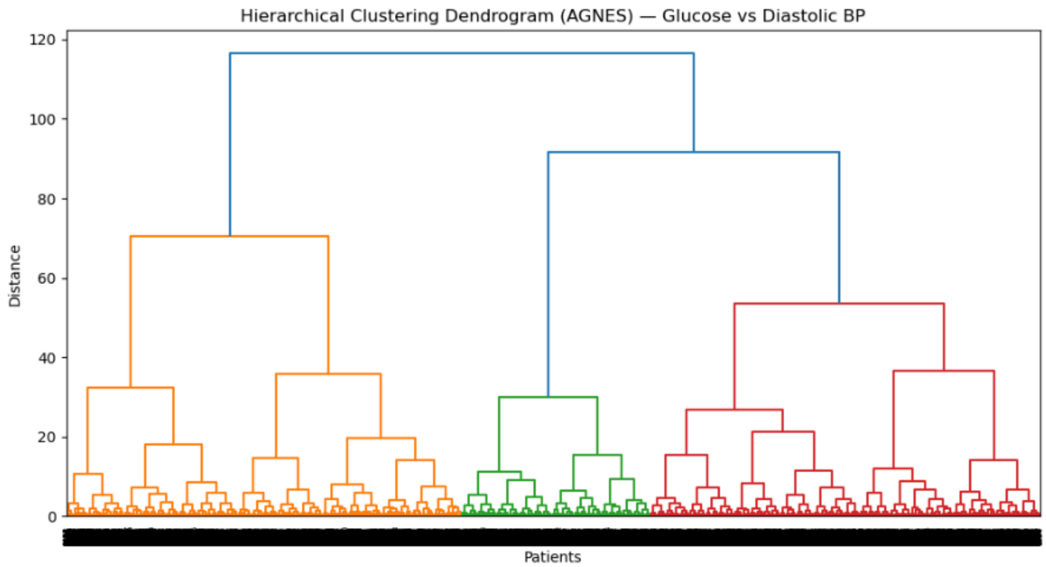
- In the K-Means clustering plot: Two scatter plot is conducted which one with Fasting_Blood_Glucose vs Blood_Pressure_Diastolic and another with Fasting_Blood_Glucose (scaled) vs Blood_Pressure_Diastolic (scaled). The first visualization illustrates the K-Means clustering before centroids were calculated and assigned points to clusters. Data points are still colored by cluster labels but at this stage, they haven't been finalized with distance calculations. So, it lacks distinct cluster centers and appears more uniformly gridded. While the second one shows the clustering after the algorithm completed its centroid calculation and Euclidean distance assignments. Here, the data points are clearly grouped into four clusters with each centered around its calculated centroid (marked with red crosses).
- The Agglomerative clustering plot: This scatter plot separates the data into three clusters. It shows that the clusters follow a diagonal and slightly curved separation: the red cluster groups individuals with lower fasting blood glucose across all diastolic blood pressure levels, the blue cluster appears at higher glucose and higher pressure levels while the green cluster represents those with higher glucose but lower diastolic pressure.
- The dendrogram shows a clear separation into three main clusters, with distinct branches forming at higher distances. The clustering reflects patterns in glucose and diastolic blood pressure where patients with similar values are grouped more closely together, supporting the use of k=3 in agglomerative clustering.

These visualizations help reveal that patients with higher Fasting Blood Glucose levels often also present with higher Blood Pressure Diastolic values.






### 3.2.4 Business Insight

This clustering analysis provides a valuable business insight: patients with elevated fasting glucose levels often exhibit higher diastolic blood pressure, indicating a dual risk for metabolic syndrome. Recognizing this co-occurrence enables more proactive health interventions.

By identifying this subgroup:

- Healthcare providers can design integrated monitoring programs for individuals with both high glucose and blood pressure readings.
- Preventive strategies can focus on lifestyle changes addressing both glucose control and cardiovascular health.
- Conversely, patients with normal glucose and blood pressure can be managed through standard care and periodic checkups.

This segmentation supports efficient resource allocation, enabling high-risk individuals to receive targeted care while maintaining cost-effective management for lower-risk populations.

### 3.3 Patients with higher HbA1c values tend to have higher Fasting_Blood_Glucose levels

As part of this analysis, an unsupervised learning technique was utilized to group patients based on their HbA1c and fasting blood glucose values. These are the two indicators of blood sugar regulation. These attributes were selected due to their strong clinical relevance in identifying individuals at risk of diabetes and related to metabolic conditions. The purpose of the clustering was to explore natural groupings among patients that might correspond to different levels of blood sugar control. Recognizing these patterns allow healthcare practitioners to better understand population health trends and tailor their responses. This approach contributes to a more focused and effective method of patient management, ensuring that resources are directed where they are most needed.

#### 3.3.1 Data Preprocessing

1. **Feature Selection**
   This line of code performs feature selection by extracting only the ‘HbA1c’ and ‘Fasting_Blood_Glucose’ columns from the dataset, which are the relevant attributes used for clustering analysis based on the relationship between long-term blood sugar levels and fasting glucose levels.

```
# Feature Selection
selected_features = df[['HbA1c', 'Fasting_Blood_Glucose']]
```

2. **Handling Missing Values**
   This line handles missing values by removing any rows from the selected features (HbA1c and Fasting_Blood_Glucose) that contain Nan values, ensuring the clustering algorithm only processes complete and valid data.

```
# Handling Missing Values
selected_features = selected_features.dropna()
```

3. **Normalization**
   This code performs normalization using StandardScaler, which standardizes the selected features (HbA1c and Fasting_Blood_Glucose) by scaling them to have a mean of 0 and a standard deviation of 1, ensuring both features contribute equally to the clustering process.

```
# Normalization
scaler = StandardScaler()
scaled_features = scaler.fit_transform(selected_features)
```

### 3.3.2 Model Evaluation

The Silhouette Score was used to assess the performance of the clustering algorithms. The Silhouette Score measures how well each data point fits within its assigned cluster compared to other clusters, with values ranging from -1 to 1. A higher score indicates better-defined clusters. For this analysis, k was set to 3 for both K-Means and AGNES. This value was chosen to reflect a moderate segmentation of patient profiles based on their health indicators. The scores provided a quantitative basis to compare the clustering quality of both algorithms using the same number of clusters. The Figure shows the result of Silhouette Score for both algorithms.

```
K-Means Silhouette Score: 0.374
AGNES Silhouette Score: 0.360
```

*Figure : Silhouette Score Results*

### 3.3.3 Visualization and Interpretation

This section presents the visual output from clustering analysis using two unsupervised learning techniques, which are K-Means and Agglomerative Hierarchical Clustering (AGNES) based on the features of HbA1c and Fasting Blood Glucose. These visualizations help to illustrate how patients are grouped into clusters with similar blood sugar characteristics, allowing us to identify patterns that may correspond to varying levels of glycemic control. By comparing the dendrogram and the cluster scatter plots, we can better understand the structure of the data and how each algorithm segments the patient population.

- K-Means Clustering Scatter Plot - This scatter plot displays the output of the K-Means algorithm applied to the features HbA1c and Fasting Blood Glucose. Each point represents a patient, coloured according to the cluster assigned by K-Means. The chart shows three distinct groups :
  - Cluster 0 : Lower HbA1c and moderate glucose levels
  - Cluster 1 : Higher HbA1c and glucose levels
  - Cluster 2 : Moderate HbA1c with high or low glucose

  The clear triangular boundary between clusters reflects how K-Means partitions the data based on distance to cluster centroids. This clustering offers a clean segmentation, although it assumes clusters to be spherical and equal in size.
- AGNES Clustering Scatter Plot - This plot shows the result of the AGNES clustering, also using 3 clusters but with a different structure than K-Means. AGNES captures more organic and irregular boundaries, better reflecting real-world variability in medical data.

The clusters here are shaped less symmetrically and the boundaries adapt to the data density and distribution, which can be helpful when patient data does not naturally form clean geometric groups.

- Hierarchical Clustering Dendrogram (AGNES) - The dendrogram shows the hierarchical structure produced by AGNES. Each leaf at the bottom represents a patient and branches represent how clusters were formed by merging the closest pairs. The vertical axis indicates the distance between clusters when they were joined. By drawing a horizontal line at a chosen height (cut-off point), we can determine the number of final clusters. In this case, 3 clusters were formed. This visual gives insight into how distinct or overlapping the patient groupings are.

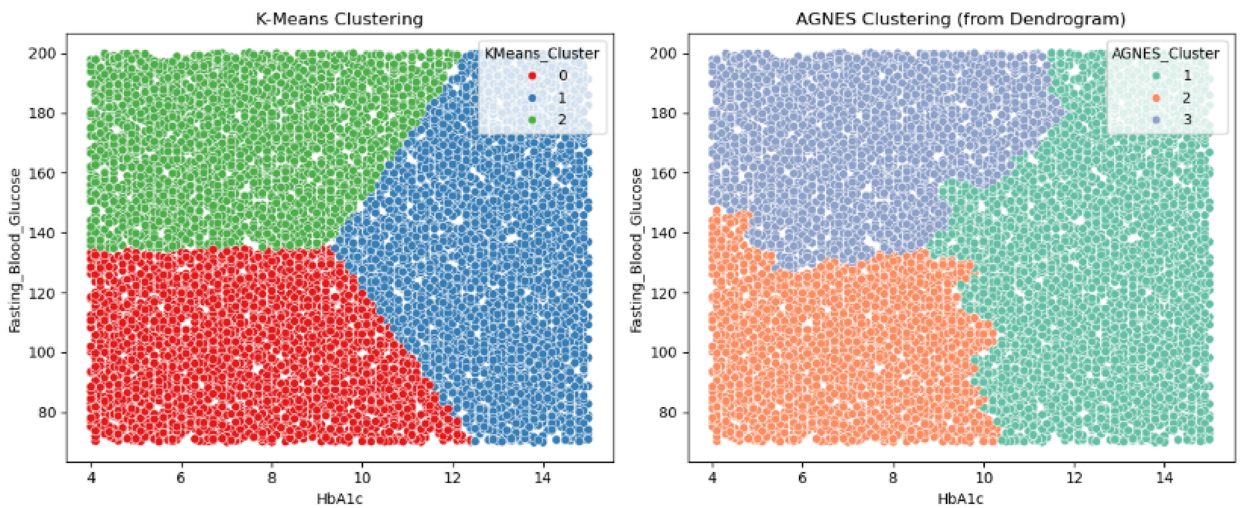



*Figure : Scatter Plots*

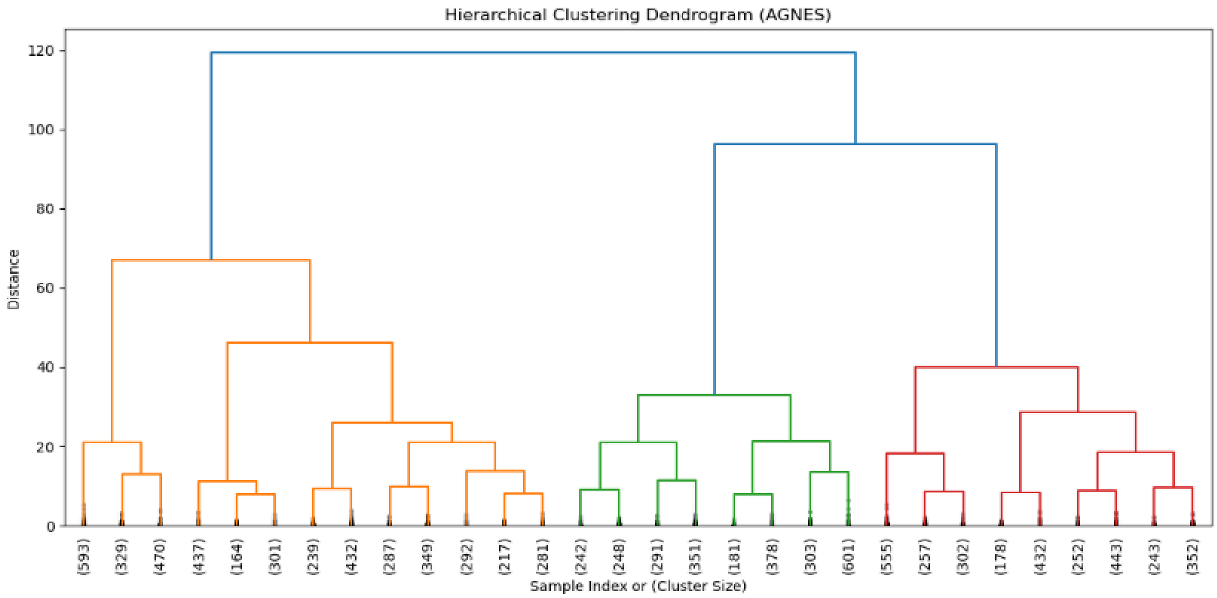



*Figure : Dendrogram*

### 3.3.4 Business Insight

This clustering analysis reveals a significant business reality. Those patients with a greater percentage of HbA1c have higher levels of fasting blood glucose and this indicates an agreement of faulty glycemic control. This shared presentation implies greater risk of diabetes progression along with chronic metabolic distress.

By identifying this subgroup:

- Physicians are able to devise referral diabetes screening and early treatment protocols for those individuals who have an abnormality on both markers
- Treatment and education plans can emphasize longer-term blood sugar control, attention to both short-term and longer-term glucose handling.
- Others with tightly controlled HbA1c and fasting blood glucose levels can merely stay with regular monitoring with standard health maintenance regimens.

This stratification permits more precise and proactive care strategies to differentiate those at highest risk and target them, optimizing resource use in the overall population.

## 4.0 CONCLUSION

In this project report, unsupervised learning techniques have been employed for the purpose of discovering meaningful patterns within health datasets of patients. The analysis using the clustering algorithms was able to create segments based on substantial clinical parameters such as BMI, cholesterol, fasting glucose, blood pressure and HbA1c. Each segment provides insights into the health risks and allows for the identification of high-risk populations. Such findings are valuable for medical practitioners to offer more tailored, precise interventions and optimize medical resources. Overall, the project demonstrates the power of data mining for advancing preventive health strategies at a higher level and improving patient chances through data-driven alias decision-making.

**SECTION B**

**REFLECTION SAFIYA NURSYAHADAH**

PART A : PROJECT REFLECTION

In Project 1, the primary focus was on supervised learning through the development of the classification models using a dataset from a previous assignment. The process started with essential data preparation, including handling missing values and selecting appropriate features for model training. One of the key techniques that I implemented in this project was the Naive Bayes algorithm, which was chosen for its simplicity and efficiency in handling classification tasks. Each model aimed to uncover specific business insights, and the results were evaluated using metrics such as accuracy, recall, and precision. These evaluations helped us compare model performances and determine which algorithm produced the most reliable predictions.

Project 2 shifted towards unsupervised learning by applying clustering algorithms to the same dataset used in project 1. I selected K-Means and Agglomerative Clustering to explore hidden patterns and group similarities within the data. Preprocessing remained a critical step, involving normalization and careful feature selection to ensure meaningful cluster formation. In this project, I specifically focused on deriving business insights for Clinical Risk Segmentation: BMI and Cholesterol Analysis, which helped me understand health related risk patterns among individuals. Visual tools like scatter plots and silhouette scores were used to interpret the results and draw relevant business insights from the clustering outcomes.

Both projects came with their own set of challenges. In Project 1, balancing model complexity and performance was crucial, while in Project 2, interpreting unlabeled clusters posed difficulty. However, these experiences significantly enhanced my understanding of practical data mining. I developed stronger skills in data analysis, model evaluation, and communicating technical results. Working in a team also improved my collaboration and report writing skills, which are valuable for future academic and professional work.

**PART B : COURSE REFLECTION**

This course offered a comprehensive overview of data mining concepts, techniques, and real world applications. From the initial introduction discovery and CRISP-DM, to hands-on projects involving classification and clustering. I appreciated the balance between theoretical understanding and practical tasks, especially in the project work.

Reflecting on the intended learning outcomes, I believe I have achieved the goals set for this course. I am now confident in analyzing data, applying appropriate techniques, and interpreting the results critically. The structures approach and step-by-step project guidance helped me overcome initial difficulties, especially in the clustering topics. Though unsupervised learning required deeper analysis, it improved my ability to work with complex, unlabeled data.

Throughout the course, I gained technical data and transferable skills, including data preprocessing, model building, performance evaluation, and teamwork. These skills are not only relevant to academic projects but are also essential in professional data driven environments. I intend to apply what I have learned in future research especially in tasks involving data analysis, report writing, and decision making support through data insights.

**REFLECTION FARRA NURZAHIN**

**PART A: PROJECT REFLECTION**

In project 1, I focused on supervised learning which is classifying to create and test using classification models which is Support Vector Machine (SVM). Specifically, I employed 3 business insights in the models which are that moderate physical activity is connected to improved blood sugar control, males have a little higher BMI than females and heavy drinkers exhibit increased. The process included particular data cleaning, addressing missing values, encoding categorical variables, stratified sampling to balance classes and lastly training and testing the models. In contrast, Project 2, I explored unsupervised learning by employing K-Means and Agglomerative Hierarchical Clustering including Agglomerative Nested (AGNES) to discover natural clustering within the data. We looked for patterns that linked BMI to cholesterol, fasting glucose to diastolic blood pressure and HbA1c to fasting glucose. The goal was to identify meaningful clusters and reveal patterns that might not be immediately visible through supervised learning. Key patterns explored included the connection between fasting glucose and diastolic blood pressure. Furthermore, I added informative visualization like scatter plot and dendrograms to illustrate the distribution of key variables and to visually validate the relationship in the business insights. These visualizations were essential in making a complex dataset understandable.

During these tasks, we experienced a few obstacles. In project 1, it is difficult to handle unbalanced data particularly for multi-class classification tasks such as HbA1c category prediction, where some classes had very few samples. For project 2, determining the right number of clusters took careful review with silhouette scores as well as interpreting the clustering boundaries. Despite such obstacles, both projects helped me understand the machine learning process, from data preprocessing to model selection and evaluation. Working on these projects has greatly enhanced our technical and analytical skills. We also learned how to evaluate model performance using measures such as precision, recall, F1-score and silhouette score as well as how to transform technical discoveries into actionable insights that can help with better planning.

**PART B: COURSE REFLECTION**

Overall, this course provided a practical and thorough introduction to data mining and machine learning which combines theoretical and understanding with practical experience. These projects gave me the experience to conduct supervised and unsupervised learning on real data which makes the concepts I learned in lecturers much clearer and more useful. Furthermore, I feel that I met the majority of the planned learning outcomes, especially in terms of data preprocessing techniques, model building and comparison as well as extracting business insights from the dataset.

The new skills I have learned are to improve my coding and analytical skills by becoming more comfortable with Python libraries and tools utilized in the data science workflow. I also learn how to turn datasets into useful insights which is an important ability for future work in data-driven sectors. Looking ahead, I intend to use what I’ve learned to future academic and professional efforts such in health analytics, marketing and operations. Moreover with the skill that I’ve learned, I will effectively use it to convey findings wherever data is used in decision making.

**REFLECTION DAYANG FARAH FARZANA**

**PART A : PROJECT REFLECTION**

In Project 1, I used classification using the decision tree algorithm and created a decision table to show how specific health attributes influenced the model’s prediction. I performed the following data preprocessing like handling missing values and feature selection, as well as model performance measurement by accuracy, recall and precision. The decision table offered improved comprehension of the logic behind the model and in explaining the outcome of non-experts. A major concern I had was choosing the most suitable model in which performance and explainability were balanced equally well, especially when Decision Tree was compared to complex models like SVM

In Project 2, I performed the third business insight and examined the relationship between HbA1c and Fasting Blood Glucose. I did feature selection, removed rows with missing data and applied StandardScaler to scale the data prior to which I ran K-Means and Agglomerative Hierarchical Clustering (AGNES) to cluster patients. I applied silhouette scores to determine the number of clusters that is optimal and presented the results using scatter plots and dendrograms.

One of the key challenges in this project was interpreting unlabelled clustering results, whereby it took attention to detail to unlock significant patterns. I was able to gain hands-on experience in preprocessing, building models, cluster analysis and performance measurement from both projects. I was also able to learn and appreciate more how to extract useful insights from both supervised and unsupervised machine learning models and importantly, on presenting technical outcomes appropriately within healthcare.

**PART B : COURSE REFLECTION**

This course has been a very enriching and useful experience for me. I learned new things, from the basic principles of data mining to advanced techniques like classification, clustering and association rules. I also got to know about the steps involved in finding valuable knowledge within data and how data can be transformed into information to aid in real-life decisions. The directions were easy to follow and comprehend and the lab tasks facilitated me to better understand the topics.

I gained much new information from this course. I was instructed on how to clean and preprocess data, deal with missing or unbalanced data and analyze data using certain tools. Besides technical skills, I also developed my soft skills. Working with others helped me to understand how to communicate, prepare a decent report and present our findings properly.

I believe that I have learned a lot from this course and become better in so many aspects. Initially, I was unfamiliar with much about data mining but now I can use what I have learned with greater confidence. Even though some parts were difficult, I was able to overcome them by practicing and asking questions as needed.

In the future, I will use these skills in my studies and also in my professional career. In studies, I can use the concepts of data mining in final year projects or in any research related to data. Professionally, these skills are extremely useful for careers related to data analysis, business intelligence or technology. I believe this course has given me a good foundation that will be helpful for me in many different ways.